Data Communication & Computer Networks (CN)

Topic 4 of 7 : Networking Devices

Curation in progress — 33 auto-extracted questions pending verification (see Pending Review at the end)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** verified MCQs + **33** pending review

Question Type: MCQ — every question has 4 options (A, B, C, D)

Languages: **Bilingual:** English + Hindi (हिन्दी)

Every question includes: ✔ Correct answer (why it is RIGHT) • ★ Nukta / key point (exam tip) • ✖ why each wrong option is WRONG

Verification: Every answer cross-verified **4 times** (Forouzan facts + official answer keys of Subject Test / Unit Test Solutions where available)

Syllabus Coverage — sub-topic wise order

Sub-topic curation in progress — auto-extracted questions are listed in the **"Pending Review"** section at the end.

How to read this PDF: Each question shows 4 options (A-D) in English and Hindi. The line starting with ✔ gives the correct answer and why it is right. The line starting with ★ is the **Nukta** — the key point to remember for the exam. The lines starting with ✖ explain why every wrong option is wrong. Read the question, try it yourself, then check the answer.

System: This PDF is generated by **tools/cn_build_topic.py** from **tools/data/topic4_data.py**. To add more questions, drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions are auto-classified into topics and appended here (as pending) for review. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

These questions were auto-extracted & classified from uploaded PDFs. Answers/nukte to be verified before counting.

⚠ NEW — auto-ingested, answer needs verification (pending review)

P1. Which device reduces network collision? collision

- **(A)** Hub
- **(B)** Switch
- **(C)** Repeater
- **(D)** Modem

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P2. Which device creates a separate broadcast domain? broadcast domain

- **(A)** Switch
- **(B)** Bridge
- **(C)** Router
- **(D)** Repeater

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P3. Which device stores MAC addresses in a CAM table? - MAC addresses CAM

- **(A)** Hub
- **(B)** Router
- **(C)** Switch
- **(D)** Modem

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P4. A switch receives a frame with a destination MAC address not present in its MAC table. The switch is in a network connected to a router that separates broadcast domains. What will the switch do? MAC address MAC broadcast domains

- **(A)** Send frame to router only
- **(B)** Flood in the same broadcast domain
- **(C)** Drop the frame
- **(D)** Broadcast in entire network

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P5. Router is connected between two switches. If a broadcast frame is sent in Switch-A, what will Switch-B receive? Switch A broadcast , Switch-B

(A) Broadcast
(B) Unicast
(C) Nothing
(D) Multicast

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P6. Which device will break collision domains but NOT broadcast domains? collision domain broadcast domain

(A) Switch
(B) Router
(C) Hub
(D) Gateway

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P7. A network uses VLANs and a router-on- a-stick configuration. Which device handles VLAN tagging and which device handles inter-VLAN routing? VLAN router-on-a-stick VLAN - inter VLAN routing -

(A) Switch – VLAN tags, Router – routing
(B) Router – VLAN tags, Switch – routing
(C) Both Switch and Router do VLAN tagging
(D) None do VLAN tagging

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P8. A packet travels from a wired PC → Switch → Access Point → Wireless laptop. Which devices operate at Layer-2 only? wired PC → Switch → Access Point → Wireless laptop Layer-2

(A) Switch only
(B) Access Point only
(C) Both Switch and Access Point
(D) None

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P9. A switch has 4 VLANs configured. How many broadcast domains exist? 4 VLAN broadcast domain

(A) 1
(B) 2
(C) 4
(D) Depends on router

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P10. @A B A B Perfect Match B

(A) Packet Filtering 1.Transport Layer
(B) Stateful Inspection 2.Network Layer
(C) Circuit Level 3. Application Gateway Layer Security
(D) Stateful Inspection 4. Session Layer Firewall (a) a-1, b-2, c-3, d-4 (b) a-2, b-1, c-4, d-3 (c) a-4, b-3, c-2, d-1 (d) a-3, b-1, c-4, d-2

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P11. Which type of firewall inspects packets at the network layer and decides based on IP addresses? IP { P

(A) Packet Filtering Firewall
(B) Stateful Firewall
(C) Proxy Firewall
(D) Application Firewall

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P12. Which firewall acts as an intermediary between the client and server at the application layer? } ~ P •r 2 B P

(A) Packet Filtering Firewall
(B) Stateful Firewall
(C) Proxy Firewall
(D) Circuit-level Gateway

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P13. Which of the following statements are TRUE? k l B - ƒ u? 1. Switch reduces collision domain. 2. Hub creates a separate collision domain for each port. @ƒ P 3.Router separates broadcast domains. ˆ q

(A) 1
(B) 1 3
(C) 2 3
(D) 1, 2 3

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P14. Which device operates at the Physical layer

(A) Switch
(B) Hub
(C) Router
(D) Bridge

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P15. Transport layer multiplexing allows: E A f g :h R

(A) Multiple hosts to share one IP
(B) Multiple processes to share one network connection
(C) Multiple routers to share routing table
(D) Domain name resolution

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P16. A router performs which of the following? 9 7 - A ?** (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

(A) 1,2,4
(B) 2,3,4
(C) 1,3 only
(D) 1,2,3,4

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P17. Consider the following: S1: TTL field decreases at router. S2: MAC address changes at each hop. 7: S1: TTL i S2: MAC E

(A) Only S1
(B) Only S2
(C) Both S1 and S2
(D) Neither S1 nor S2

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P18. A router drops packets due to congestion. What is this technique called? R O?**

(A) Delay policing
(B) Redundancy
(C) Random early detection
(D) Reverse flooding

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P19. What does MAC address represent? MAC E A ?**

(A) Location of device in network
(B) Unique identity of NIC
(C) Internet connectivity
(D) Routing path

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P20. Which device separates collision domains

(A) HUB
(B) Switch
(C) Repeater
(D) Bridge

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P21. What is a subnet mask used for?Subnet

(A) Encrypt data
(B) Identify default gateway
(C) Identify network and host portion
(D) Manage routing table

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P22. Which of the following devices works only on layer 1

- **(A)** Router
- **(B)** Hub
- **(C)** Switch
- **(D)** Bridge

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P23. Which device operates at all 7 layers of OSI model? OSI 7

- **(A)** Router
- **(B)** Hub
- **(C)** Gateway
- **(D)** Switch

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P24. BGP is mainly used for which type of routing?BGP

- **(A)** Intra-AS routing
- **(B)** Inter-AS routing
- **(C)** Static routing
- **(D)** Host routing

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P25. What kind of NAT maps one private IP to one public IP? - NAT IP IP

- **(A)** Dynamic NAT
- **(B)** Static NAT
- **(C)** PAT
- **(D)** Port Forwarding

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P26. What is the function of NAT in computer networks? NAT

- **(A)** Port forwarding
- **(B)** IP address translation
- **(C)** Bandwidth control
- **(D)** DNS resolution

[Source: CN Topic Test 5.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P27. MPLS operates between which two layers?MPLS

(A) Layer 1 and 2
(B) Layer 2 and 3
(C) Layer 3 and 4
(D) Layer 4 and 7

[Source: CN Topic Test 5.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P28. In specific, if the systems use separate protocols, which one of the following devices is used to link two systems

(A) Repeater
(B) Gateway
(C) Bridge
(D) Hub

[Source: CN Topic Test-01.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P29. Which of the following devices is not a networking device

(A) Hub
(B) Switch
(C) Bridge
(D) None of the these

[Source: CN Topic Test-01.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P30. Which of the following layer protocols are responsible for user and the application programme support such as passwords, resource sharing, file transfer and network management

(A) Layer 2 protocols
(B) Layer 5 protocols
(C) Layer 7 protocols
(D) Layer 10 protocols

[Source: CN Topic Test-02.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P31. Which device connects different networks and routes data between them

(A) Hub
(B) Switch
(C) Router
(D) Bridge

[Source: CN Topic Test-03.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P32. A network ID cannot be assigned to:

(A) A router
(B) A switch
(C) A host device
(D) The entire network

[Source: CN Topic Test-03.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P33. What is the primary function of the MAC sublayer? MAC

(A) Routing
(B) Error checking
(C) Addressing
(D) Channel access

[Source: CN Topic Test-04.pdf | auto-classified: Topic 4 / Router / Brouter]